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Supplementary Materials for

Deep generative models design mRNA sequences with enhanced translational capacity and stability

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MDAR Reproducibility Checklist
Data S1 to S4

Supplementary Text

Design space of full-length mRNA

For both 5′ and 3′UTRs, it is free to select one of the four nucleotides at each position; for CDS, although constrained by the specified amino acid in the given protein sequence, each codon slot has up to six choices (**Fig. 1A**). Therefore, the full-length mRNA design space *D* for a given target protein $\boldsymbol{y}$ can be formalized as:

$$D(\boldsymbol{y}) = 4^{N_1} \cdot 4^{N_2} \cdot \prod_{i=1}^{L} C(\boldsymbol{y}_i)$$

where N_1, N_2 and L are the lengths of 5′UTR, 3′UTR and the target protein, respectively, and $C(\boldsymbol{y_i})$ is a function mapping from the *i*-th amino acid in $\boldsymbol{y}$ to the number of its possible codons. Typically, for therapeutic mRNAs, N_1 and N_2 range from 40 to 200 *nt*, and *m* varies from a hundred to more than a thousand of amino acids, making the whole design space possibly greater than 10^{500}.

Casual language modeling of GEMORNA

GEMORNA comprises two independent modules, GEMORNA-CDS and GEMORNA-UTR, both utilizing causal language modeling, a training approach focused on predicting the next tokens in a sequence. For GEMORNA-CDS, the focus is on designing CDS sequences based on a given protein sequence. Let $y = (y_1, y_2, \dots, y_L)$ represent a sequence of amino acids specifying a protein, and $x = (x_1, x_2, \dots, x_L)$ denote the corresponding set of associated codons, where L specifies the length of the protein sequence. The training objective of CDS design is to model the conditional probability $P(x|y)$, where the goal is to predict the most probable CDS sequence x given the protein sequence y. Specifically, we can train a neural network with parameters to minimize the negative log-likelihood over the entire training dataset $D_c = \{(x,y)^1, (x,y)^2 \dots, (x,y)^N\}$

$$L(D_c) = -\frac{1}{N}\sum_{i=1}^{N}\sum_{t=1}^{L} \log p_\theta\left(x_t^i \mid x_{<t}^i, y^i\right)$$

where $p_\theta\left(x_t^i \mid x_{<t}^i, y^i\right)$ represents the probability of the *t*-th codon for *i*-th instance given the preceding codons and the entire protein sequence y^i.

To achieve this, GEMORNA-CDS employs a transformer architecture (*22*). Firstly, the protein sequence is tokenized by considering each amino acid as an individual token. The sequence's start and end are augmented with "sos" (start of sequence) and "eos" (end of sequence), respectively. To align the lengths of the protein and mRNA coding sequences, the mRNA sequence is tokenized with codons. The absolute positional embeddings are then combined with the learnable token embeddings on an element-wise basis to serve as the input for the model.

The encoder processes the protein sequence, capturing intra-sequence relationships using self-attention mechanisms. This enables each amino acid to attend to other ones, efficiently modeling long-range dependencies in the sequence. The encoder outputs contextual representations that are passed to the decoder later. In decoder, the codon vectors first pass through a masked self-attention layer, ensuring that the prediction of each codon x_t^i depends only on the preceding codons $x_1^i, x_2^i, \dots, x_{t-1}^i$. These embeddings are then integrated with the encoder's context vectors through cross-attention, enabling the decoder to align the generated codons with the input amino acid sequence while preserving the essential constraints of codon selection.

For GEMORNA-UTR, the focus is on designing UTR sequences through causal language modeling. Unlike GEMORNA-CDS, which employs an encoder-decoder architecture, GEMORNA-UTR uses a standalone decoder module to predict each token in the UTR sequence based on the preceding nucleotides. $u = (u_1, u_2, \dots, u_L)$ represents a UTR sequence with L tokens. The training objective is to minimize the negative log-likelihood over the training dataset $D_u = \{u^1, u^2 \dots, u^N\}$

$$L(D_u) = -\frac{1}{N}\sum_{i=1}^{N}\sum_{t=1}^{L}\log p_\theta\left(u_t^i \mid u_{<t}^i\right)$$

The UTRs are tokenized into *k*-mers (e.g., 3-mers) and augmented with "sos" and "eos" tokens. In GEMORNA-UTR, tokens are represented as learnable embeddings. GEMORNA-UTR employs a GPT-like decoder model (*56*), focusing on autoregressive generation for UTR sequence.

Decoding strategies of GEMORNA

GEMORNA employs multiple decoding strategies to generate mRNA sequences, with all autoregressive generation processes starting from the "sos" token. For GEMORNA-CDS, three distinct methods are incorporated. The greedy search selects the token with the highest probability at each step as the next token.

$$x_t = \text{argmax} \log p_\theta(x_t \mid x_{<t}, y)$$

This approach generates deterministic sequences with minimal variation, aligning closely with the highest-confidence predictions of the model.

Beam search maintains the top-*b* most probable sequence candidates at each step, scoring them based on cumulative probabilities. At every decoding step, all possible extensions of the current candidates are considered, and the *b* best sequences are retained. This strategy considers both the current token probabilities and the cumulative prefix score.

Additionally, we investigate sampling-based decoding, which introduces stochasticity by selecting tokens based on the probability distribution at each step. This strategy enables greater exploration of the sequence design space, reflecting variability observed in natural sequences. For CDS generation, we experimented with "biased sampling", where the original probabilities are exponentiated to a specified power and re-normalized. This operation sharpens the distribution, amplifying the differences between codons and favoring those with higher probabilities. For UTR generation, we employed sampling-based decoding, promoting diversity while maintaining sequence distributions closely aligned with natural sequences.

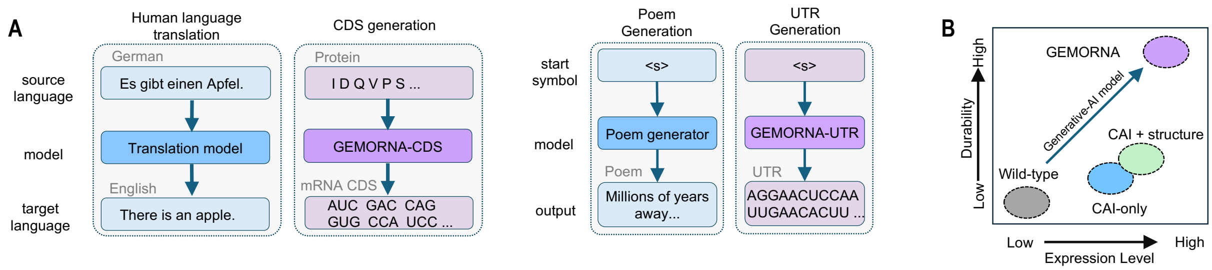

Fig. S1. Overview of GEMORNA models in the perspective of sequence generation tasks and a mRNA optimization task. **(A)** Analogous to human language translation, GEMORNA-CDS transforms a protein sequence into a CDS sequence. Analogous to poem generation, GEMORNA-UTR generates UTR sequences from scratch. **(B)** GEMORNA can de novo design therapeutic mRNAs with higher expression levels and durability versus previous rational design methods, such as those that use CAI (codon adaptation index), with or without structural information.

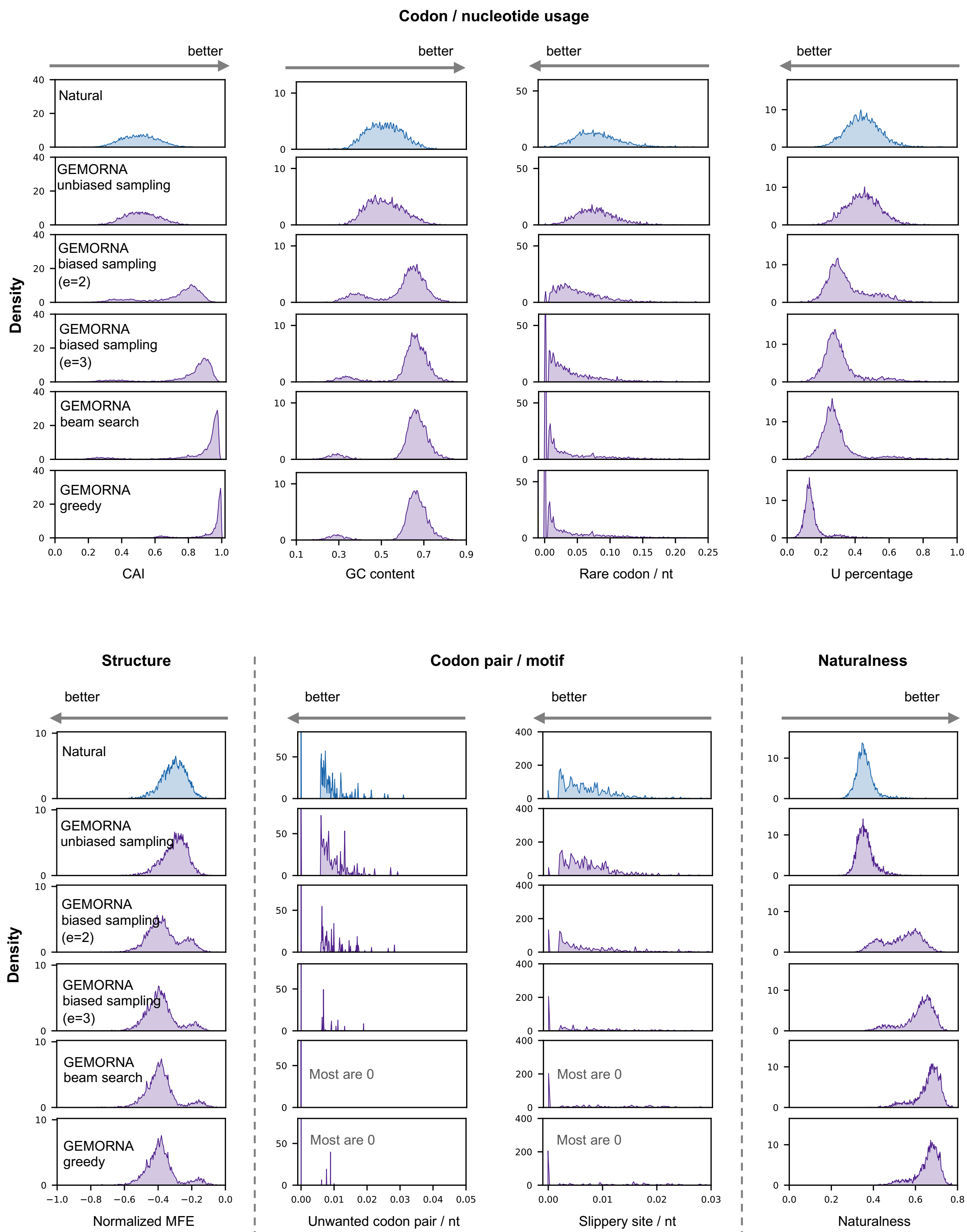

Fig. S2. One-dimensional visualizations of various optimization objectives for CDSs generated by different GEMONRA decoding strategies. “GEMORNA unbiased sampling” selects codons by sampling directly from the softmax probability distribution at each step, while “GEMORNA biased sampling” modifies this distribution by exponentiating the original probabilities to a specified power of *e*, followed by re-normalization. "GEMORNA beam search" and "GEMORNA greedy" apply classical beam search and greedy decoding methods.

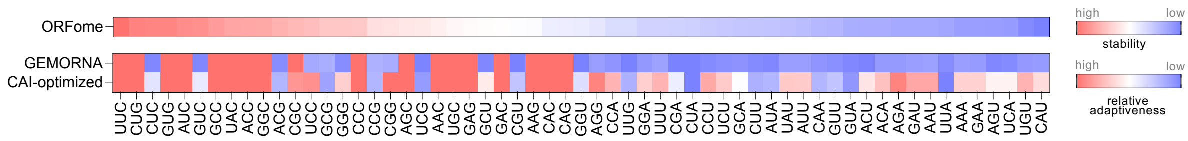

Fig. S3. A previous study investigated codon-dependent mRNA decay using an ORFome assay and used the previously described codon stability coefficient (CSC) as a measure of a codon's contribution to mRNA stability (*27*). We calculated codon relative adaptiveness derived from our GEMORNA sequences versus CAI-optimized sequences and compared them to CSC. Notably, GEMORNA-designed sequences strongly avoid unstable codons, resulting in a higher correlation with CSC.

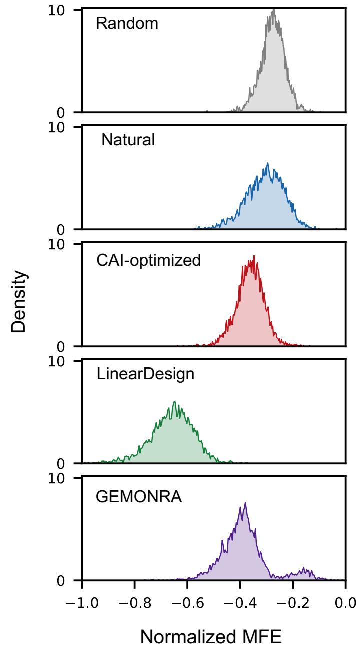

Fig. S4. The length-normalized minimum free energy (MFE) distribution of CDSs generated from different sources including LinearDesign (*13*).

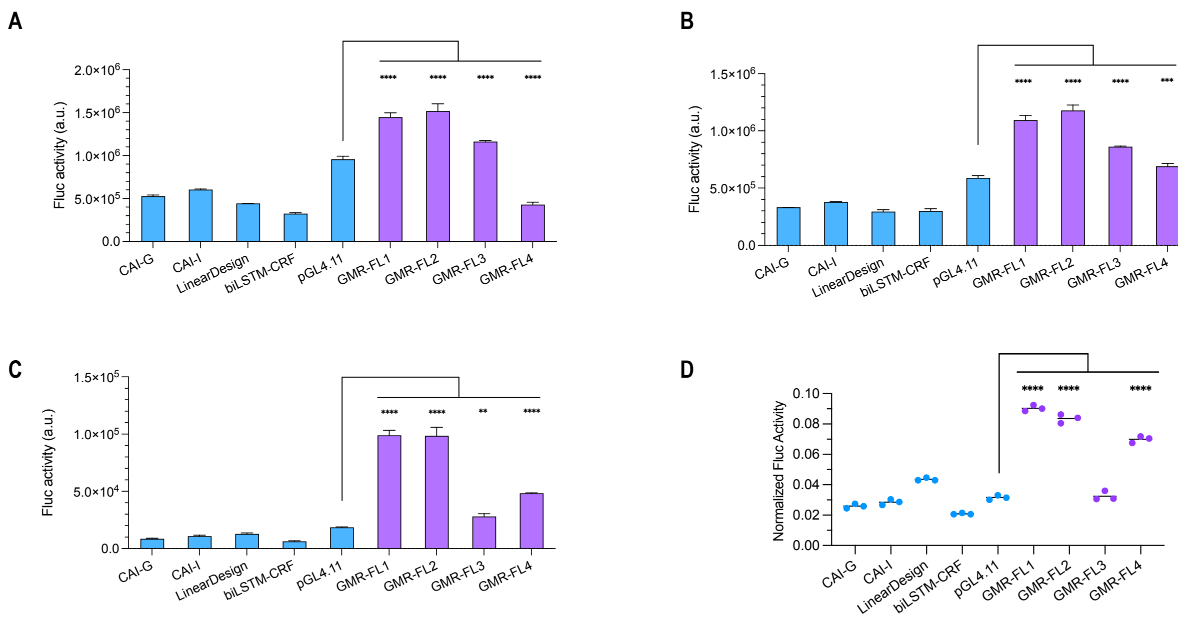

Fig. S5. More comparisons of expression levels and duration between GEMORNA CDSs and control CDSs. **(A** and **B)** Firefly luciferase (Fluc) activity measured at 24 hours post-transfection in HEK293T **(A)** and HepG2 **(B)** cells. **(C)** Fluc activity at 48 hours in HepG2 cells. **(D)** Ratio of expression levels in HepG2 cells at 48 hours relative to 24 hours. Experiments were performed with $n = 3$ biological replicates. Error bars represent the standard deviation of mean. One-way ANOVA was used for statistical analysis (**$P < 0.01$, ***$P < 0.001$, ****$P < 0.0001$).

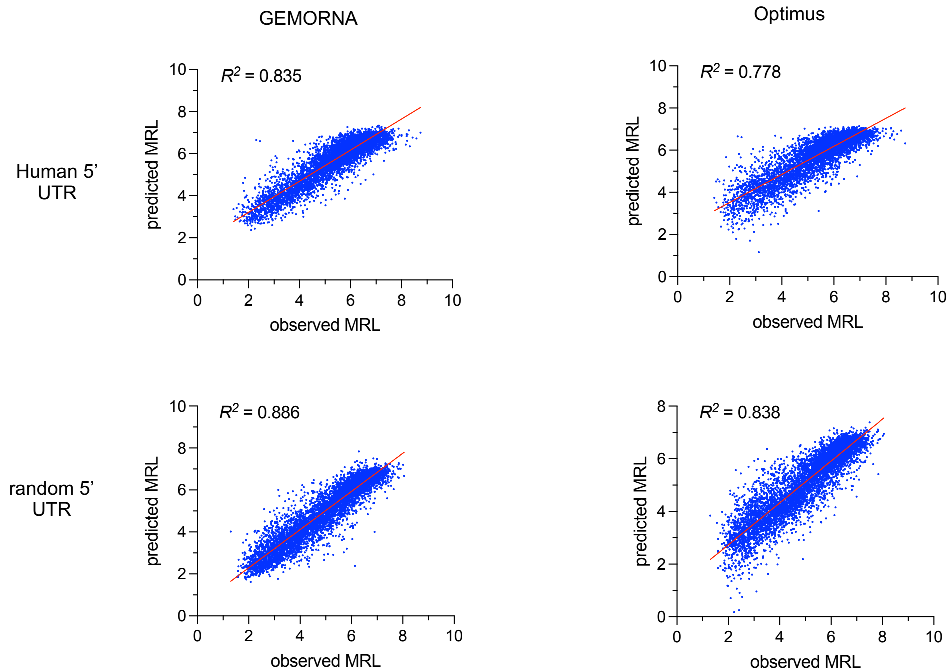

Fig. S6. Correlations between observed mean ribosome load (MRL) and predicted MRL on human 5′UTRs (first row) and random 5′UTRs (second row). The first column presents the correlations from our predictor PRED-5UTR, while the second column displays the results from Optimus (*18*).

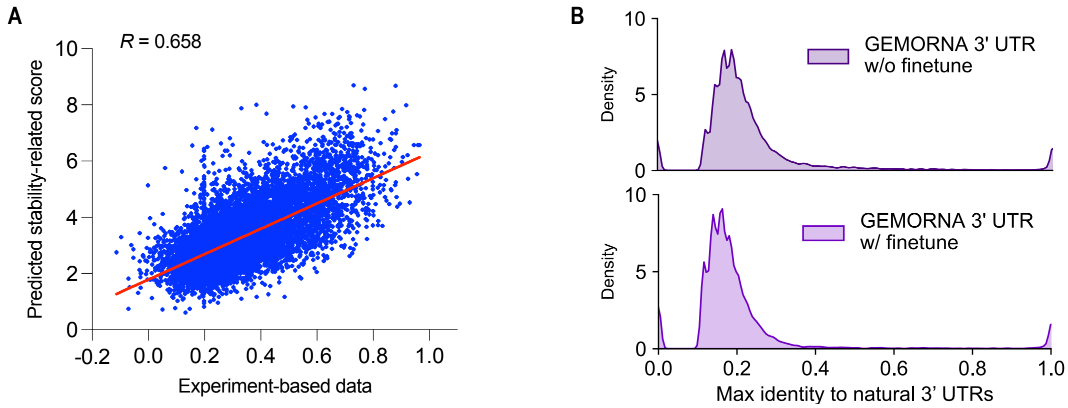

Fig. S7. *In silico* analysis of 3′UTRs. **(A)** Correlation plot of predicted 3′UTR stability-related score by PRED-3UTR model against the experimental-based data (*38*). **(B)** Max identity of GEMORNA-generated 3′UTRs compared to natural 3′UTRs.

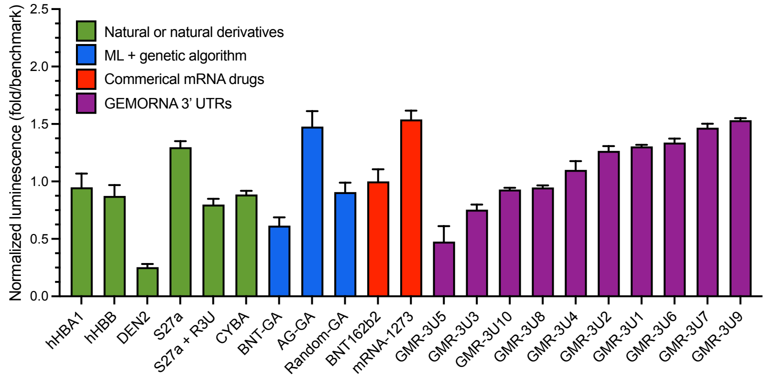

Fig. S8. Normalized firefly luminescence of GEMORNA 3′UTRs compared to various benchmarks at 48 hours in HEK293T cells. All values were normalized to BNT162b2’s 3′UTR. GMR-3U4 to GMR-3U7 were derived from GMR-3U1 by incorporating stabilized motifs, but these modifications did not result in a noticeable improvement in the expression levels. Experiments were performed with n = 3 biological replicates. Error bars represent the standard deviation of mean.

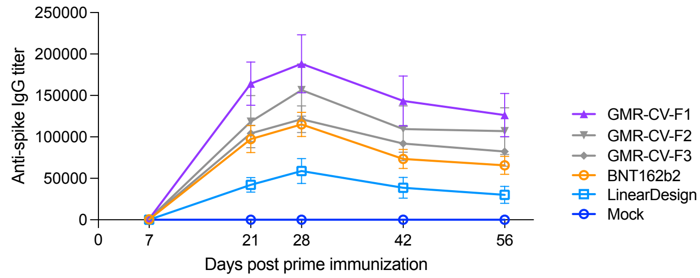

Fig. S9. *In vivo* anti-spike IgG comparisons of COVID-19 mRNA vaccine sequences at a 1 μg dosage (n = 5 mice per group). GMR-CV-F1, GMR-CV-F2 and GMR-CV-F3 represent three GEMORNA-generated full-length designs. BNT162b2 represents the mRNA sequence of Pfizer/BioNTech COVID-19 vaccine. LinearDesign represents the LinearDesign Seq-C CDS combined with UTRs from BNT162b2. For LinearDesign sequence, two corresponding codons were altered to include the "2P" mutation in protein sequence. For all sequences, m1Ψ modification were used for mRNA synthesis in place of unmodified uridine. Error bars represent the geometric error of geometric mean.

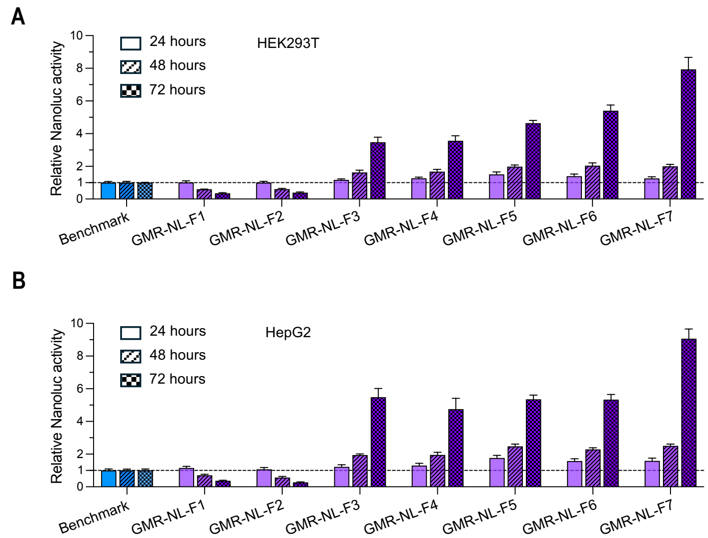

Fig. S10. Relative NanoLuc expression levels of GEMORNA mRNAs and benchmark mRNAs in HEK293T **(A)** and HepG2 **(B)** cells. Experiments were performed with n = 3 biological replicates. Error bars represent the standard deviation of mean.

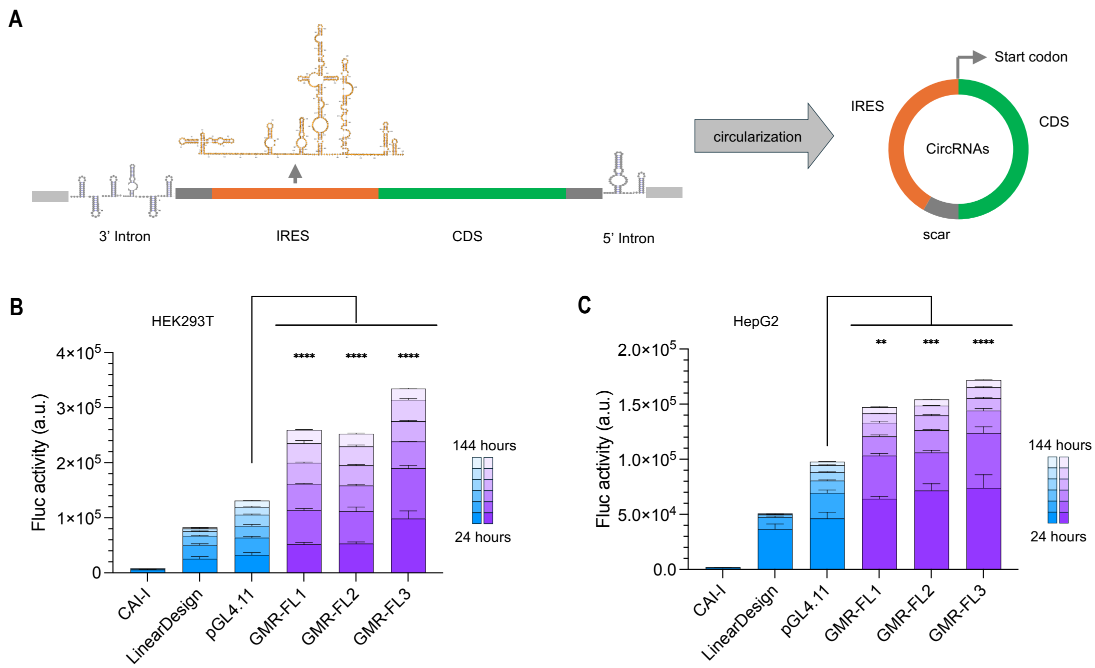

Fig. S11. The GEMORNA platform is able to generate circRNA CDSs with enhanced expression levels and durability. **(A)** Visualization of circular RNA components and circularization process. For the circRNA constructs, we followed the intron-based circularization method (23) and utilized the initial ribosome entry site (IRES) from the encephalomyocarditis virus (EMCV) genome. **(B)** and **(C)** Firefly luciferase activities in HEK293T **(B)** and HepG2 **(C)** cells. Experiments were performed with n = 3 biological replicates. Error bars represent the standard deviation of mean. One-way ANOVA was used for statistical analysis (**$P < 0.01$, ***$P < 0.001$, ****$P < 0.0001$).

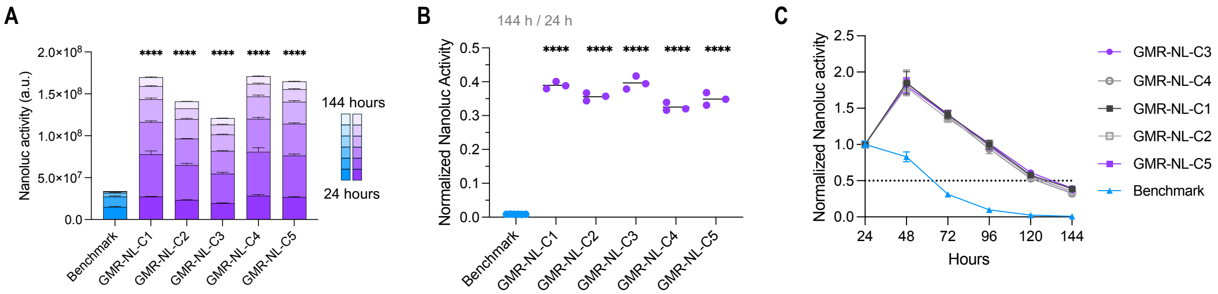

Fig. S12. Expression levels and durability of NanoLuc circRNAs in HepG2 cells. **(A)** Comparison of accumulated NanoLuc luciferase between GEMORNA circRNAs and the benchmark circRNA. NanoLuc luciferase activities were measured every 24 hours. **(B)** Ratios of NanoLuc product at 144 hours to product at 24 hours. **(C)** Normalized NanoLuc luciferase activities from 24 hours to 144 hours after transfection; the data were normalized to NanoLuc luciferase activity at 24 hours. Experiments were performed with n = 3 biological replicates. Error bars represent the standard deviation of mean. One-way ANOVA was used for statistical analysis (*****P* < 0.0001).

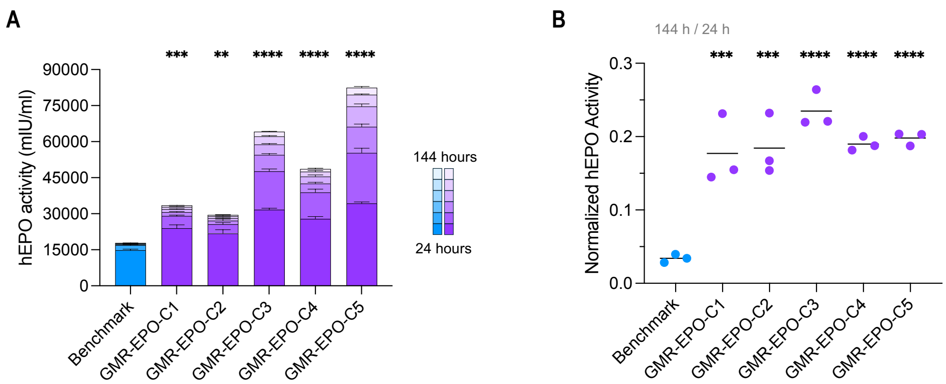

Fig. S13. Expression levels and durability of EPO circRNAs in HepG2 cells. **(A)** Comparison of accumulated EPO between GEMORNA circRNAs and the benchmark. **(B)** EPO expression level ratios of 144 hours to 24 hours. Experiments were performed with n = 3 biological replicates. Error bars represent the standard deviation of mean. One-way ANOVA was used for statistical analysis (**$P < 0.01$, ***$P < 0.001$, ****$P < 0.0001$).

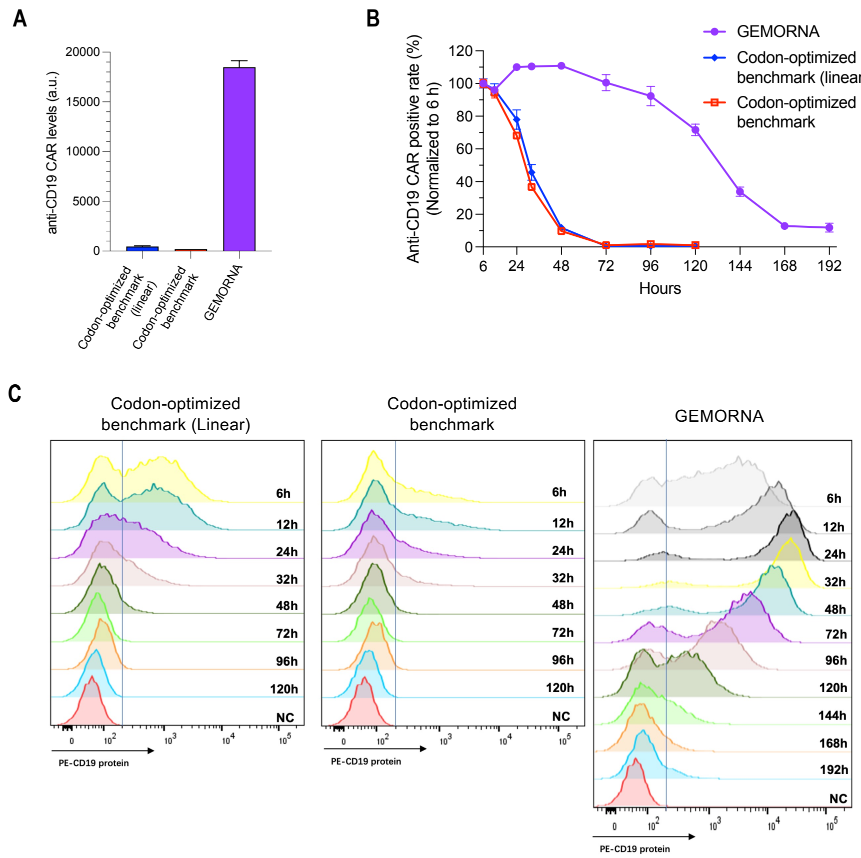

Fig. S14. Comparisons of CD19 CAR expression levels between GEMORNA circRNA and two conventionally designed benchmark RNAs. The two benchmarks have the same CDS, but one is linear RNA, denoted as "Codon-optimized benchmark (Linear)" and the other is circRNA, denoted as "Codon-optimized benchmark". **(A)** Expression levels of the three sequences at 24 hours after electro-transfected into human T cells. **(B)** Changes in the CD19 CAR positive rate over 192 hours. Electroporation efficiency was normalized by positive rate at 6 hours post electroporation. **(C)** Detailed CD19 CAR expression from 6 hours to 192 hours after electro-transfected into Jurkat cells. Experiments were performed with n = 3 biological replicates. Error bars represent the standard deviation of mean.

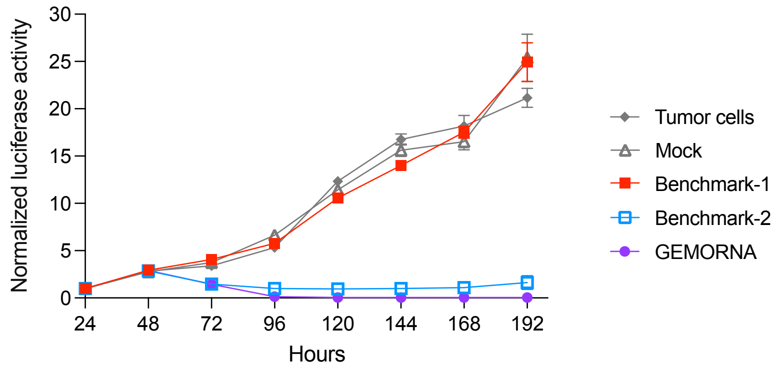

Fig. S15. T cell-mediated killing of NALM6 cells with an effector:target (E:T) ratio of E:T=1:3. "Tumor cells": no T cell-mediated treatment. "Mock": T cell-mediated treatment without CD19 CAR circRNAs transfected. "Benchmark-1", "Benchmark-2" and "GEMORNA": T cell-mediated treatments with transfected codon-optimized circRNAs, patented circRNAs and GEMORNA circRNAs, respectively; all these circRNAs encode CD19 CAR. Experiments were performed with n = 3 biological replicates. Error bars represent the standard deviation of mean.

Table S1. Detailed sequence information for mRNAs used in this study.

Figure	Name	5′UTR	CDS	3′UTR
Fig. 4B	Benchmark-1	AG	CAI-I	hHBA1
Fig. 4B	Benchmark-2	BNT162b2	pGL4.11	BNT162b2
Fig. 4B	GMR-FL-F1	GMR-5U7	GMR-FL2	GMR-3U1
Fig. 4B	GMR-FL-F2	GMR-5U7	GMR-FL2	GMR-3U2
Fig. 4B	GMR-FL-F3	GMR-5U9	GMR-FL2	GMR-3U1
Fig. 4B	GMR-FL-F4	GMR-5U9	GMR-FL2	GMR-3U2
Fig. 4B	GMR-FL-F5	GMR-5U6	GMR-FL2	GMR-3U9
Fig. 4E	BNT162b2	BNT162b2	BNT162b2	BNT162b2
Fig. 4E	LinearDesign	BNT162b2	LinearDesign (Covid-19 spike)	BNT162b2
Fig. 4E	GMR-CV-F1	GMR-5U7	GMR-CV1	GMR-3U1
Fig. 4E	GMR-CV-F2	GMR-5U7	GMR-CV1	GMR-3U2
Fig. 4E	GMR-CV-F3	GMR-5U6	GMR-CV1	GMR-3U9
Fig. 4I	Benchmark	BNT162b2	Chen et al. (EPO)	BNT162b2
Fig. 4I	GMR-EPO-F1	GMR-5U7	GMR-EPO1	GMR-3U1
Fig. 4I	GMR-EPO-F2	GMR-5U7	GMR-EPO3	GMR-3U1
Fig. 4I	GMR-EPO-F3	GMR-5U9	GMR-EPO2	GMR-3U2
Fig. 4I	GMR-EPO-F4	GMR-5U9	GMR-EPO4	GMR-3U2
Fig. 4I	GMR-EPO-F5	GMR-5U6	GMR-EPO1	GMR-3U9
Fig. 4I	GMR-EPO-F6	GMR-5U6	GMR-EPO4	GMR-3U9
Fig. 4I	GMR-EPO-F7	GMR-5U6	GMR-EPO5	GMR-3U9
Fig. S10	Benchmark	BNT162b2	Chen et al. (NanoLuc)	BNT162b2
Fig. S10	GMR-NL-F1	GMR-5U7	GMR-NL1	GMR-3U1
Fig. S10	GMR-NL-F2	GMR-5U7	GMR-NL3	GMR-3U1
Fig. S10	GMR-NL-F3	GMR-5U9	GMR-NL2	GMR-3U2
Fig. S10	GMR-NL-F4	GMR-5U9	GMR-NL4	GMR-3U2
Fig. S10	GMR-NL-F5	GMR-5U6	GMR-NL1	GMR-3U9
Fig. S10	GMR-NL-F6	GMR-5U6	GMR-NL4	GMR-3U9
Fig. S10	GMR-NL-F7	GMR-5U6	GMR-NL5	GMR-3U9

Table S2. Detailed sequence information for circular RNA used in this study.

Name	5′UTR	IRES	CDS	3′UTR
Benchmark	PABP	iHRV-B3 + Apt-eIF4G	Chen et al. (NanoLuc)	hHBA1
GMR-NL-C1		PbV-A4	GMR-NL1	
GMR-NL-C2		PbV-A3	GMR-NL1	
GMR-NL-C3		PbV-A4	GMR-NL2	
GMR-NL-C4		PbV-A4	GMR-NL3	
GMR-NL-C5		PbV-A3	GMR-NL4	
Benchmark	PABP	iHRV-B3 + Apt-eIF4G	Chen et al. (EPO)	hHBA1
GMR-EPO-C1		PbV-A4	GMR-EPO1	
GMR-EPO-C2		PbV-A3	GMR-EPO1	
GMR-EPO-C3		PbV-A4	GMR-EPO2	
GMR-EPO-C4		PbV-A4	GMR-EPO3	
GMR-EPO-C5		PbV-A3	GMR-EPO4	
Benchmark-1		CVB3	Benchmark-1 (CD19 CAR)	
Benchmark-2		CVB3	Benchmark-2 (CD19 CAR)	
GEMORNA		CVB3	GEMORNA	
CAI-I		EMCV	CAI-I	
LinearDesign		EMCV	LinearDesign (Firefly luciferase)	
pGL4.11		EMCV	pGL4.11	
GMR-FL1		EMCV	GMR-FL1	
GMR-FL2		EMCV	GMR-FL2	
GMR-FL3		EMCV	GMR-FL3	

Data S1. Sequences used in this study

Data S2. Raw data of one-dimensional and two-dimensional visualizations for CDSs

Data S3. Raw data of correlations between the optimization objectives and experimental data for CDSs

Data S4. Raw data for *in silico* analysis of GEMORNA-generated 5′UTRs

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